

Horseshoe Lemma (Projective version)

Setting:

- a short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$
- projective resolutions $A_\bullet$ and $C_\bullet$ of A and C respectively
- Within abelian category with enough projectives

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \downarrow d_2^A & & \downarrow & & \downarrow d_2^C & \\
 & A_1 & & B_1 & & C_1 & \\
 & \downarrow d_1^A & & \downarrow & & \downarrow d_1^C & \\
 & A_0 & & B_0 & & C_0 & \\
 & \downarrow \varepsilon_A & & \downarrow & & \downarrow \varepsilon_C & \\
 0 & \rightarrow A & \xrightarrow{\iota} & B & \xrightarrow{\pi} & C & \rightarrow 0 \\
 & \downarrow & & & & \downarrow & \\
 & 0 & & & & 0 &
 \end{array}$$

Statement:

There is a projective resolution $B_\bullet$ of B that induces a short exact sequence of complexes $0 \rightarrow A_\bullet \xrightarrow{\iota_\bullet} B_\bullet \xrightarrow{\pi_\bullet} C_\bullet \rightarrow 0$.

Proof:

We proceed by induction and note that both the base case and induction step involve the same outline:

④: define a module (B_0 or B_n) and maps (ε_B or d_n^B)

①: verify the maps make the appropriate squares commute

②: verify the maps maintain exactness (at B or B_n)

Base Case $n = \emptyset$:

⑥: Let $B_0 = A_0 \oplus C_0$ be our initial chain module in our prospective resolution of B . We then further define z_0 and π_0 to be the canonical inclusion and projection associated to our direct sum. For the vertical map $A_0 \oplus C_0 \rightarrow B$, we will define it using a map $A_0 \rightarrow B$ and a map $C_0 \rightarrow B$. The latter map is obtained via the projectivity of C_0 and the surjectivity of π ;

$$\begin{array}{ccccc} A_0 & \xrightarrow{\quad} & A_0 \oplus C_0 & \xrightarrow{\quad} & C_0 \\ \downarrow & & \downarrow & & \downarrow \\ A & \xrightarrow{\quad} & B & \xrightarrow{\quad} & C \end{array}$$

call it $h_0: C_0 \rightarrow B$ and note we have $\pi h_0 = \varepsilon_C$. The map from A_0

to B is simply $z_0 \varepsilon_A$, so we now define $\varepsilon_B: A_0 \oplus C_0 \rightarrow B$ acting on elements by $(a, c) \mapsto (z_0 \varepsilon_A)(a) + h_0(c)$. Since it is the sum of homomorphic outputs, h_0 is also a homomorphism.

⑦: For the commutativity of the left square we have $(\varepsilon_B \circ z_0)(a_0) = \varepsilon_B(z_0(a_0)) = \varepsilon_B(a_0, \emptyset) = (z_0 \varepsilon_A)(a) + \cancel{h_0(\emptyset)} = (z_0 \varepsilon_A)(a)$ where we used the fact that h_0 sends zero to zero because it is a homomorphism. This is exactly the left-bottom path in the diagram. For the right square the left-bottom path is $(\pi \circ \varepsilon_B)(a_0, c_0) = \pi((z_0 \varepsilon_A)(a_0) + h_0(c_0)) = (\pi \circ z_0 \varepsilon_A)(a_0) + (\pi \circ h_0)(c_0)$. By exactness, $\pi \circ z = \emptyset$, and then $\pi \circ h_0 = \varepsilon_C$ due to projectivity, thus we yield what we desired:

$$(\pi \circ \varepsilon_B)(a_0, c_0) = \varepsilon_C(c_0) = (\varepsilon_C \circ \pi_0)(a_0, c_0)$$

②: We seek to leverage the snake lemma since we have two SES's with commuting maps connecting them.

This yields the exact sequence $0 \rightarrow \ker \varepsilon_A \rightarrow \ker \varepsilon_B \rightarrow \ker \varepsilon_C \rightarrow \operatorname{coker} \varepsilon_A \rightarrow \operatorname{coker} \varepsilon_B \rightarrow \operatorname{coker} \varepsilon_C \rightarrow 0$. But since ε_A and ε_C are surjections, we get the truncated exact sequence $0 \rightarrow \operatorname{coker} \varepsilon_B \rightarrow 0$. This directly implies $\operatorname{coker} \varepsilon_B = 0$, thus ε_B is surjective. Note also that we get the following short exact sequence of kernels which we will need for induction: $0 \rightarrow \ker \varepsilon_A \rightarrow \ker \varepsilon_B \rightarrow \ker \varepsilon_C \rightarrow 0$

Induction Step:

③: Let $B_n = A_n \oplus C_n$, and further let ι_n and π_n be the canonical inclusion and projection. We form the diagram at right validly because our induction hypotheses gave us the short exact sequence of kernels.

$$\begin{array}{ccccc} A_n & \xrightarrow{\iota_n} & A_n \oplus C_n & \xrightarrow{\pi_n} & C_n \\ d_n^A \downarrow & & \downarrow & & \downarrow d_n^C \\ \ker d_{n-1}^A & \xrightarrow{\overline{\iota_{n-1}}} & \ker d_{n-1}^B & \xrightarrow{\overline{\pi_{n-1}}} & \ker d_{n-1}^C \end{array}$$

Then projectivity of C_n and surjectivity of $\overline{\pi_{n-1}}$ give us a map

$h_n: C_n \rightarrow \ker d_{n-1}^B$ with $d_n^C = \overline{\pi_{n-1}} h_n$. Note that $d_n^A(a_n) = d_n^A(a_n)$ and likewise with d_n^C , so the d_n 's are no more than our differentials with adjusted codomains. Now define $d_n^B: B_n \rightarrow B_{n-1}$ by $(a_n, b_n) \mapsto (\overline{\iota_{n-1}} d_n^A)(a_n) + h_n(b_n)$, and then define d_n^B to be exactly d_n^B with codomain adjusted to $\ker d_{n-1}^B$. This map is a homomorphism because it is the sum of homomorphic outputs. Note that we also obtain the chain condition $\operatorname{im} d_n^B \subseteq \ker d_{n-1}^B$ due to the construction of d_n^B .

□

①: Now we seek to show appropriate commutativity resulting from our above definitions. Consider the leftmost square in the diagram and the output of $d_n^B z_n$ on $a_n \in A_n$:

$$\begin{aligned} (d_n^B z_n)(a_n) &= d_n^B(a_n, \phi) = (\overline{z_{n-1}} d_n^A)(a_n) + h_n(\phi) \\ &= z_{n-1}(d_n^A(a_n)) + \phi = (z_{n-1} d_n^A)(a_n) \end{aligned}$$

We simply employed available definitions as well as the homomorphism property of h_n to say $h_n(\phi) = \phi$. For the right square we then consider the composite map $\pi_{n-1} d_n^B$ acting on $(a_n, c_n) \in B_n$:

$$\begin{aligned} (\pi_{n-1} d_n^B)(a_n, c_n) &= (\pi_{n-1} \overline{z_{n-1}} d_n^A)(a_n) + (\pi_{n-1} h_n)(c_n) = (\pi_{n-1} z_{n-1})(d_n^A(a_n)) + (\overline{\pi_{n-1}} h_n)(c_n) \\ &= \phi + d_n^C(c_n) = (d_n^C \pi_n)(a_n, c_n) \end{aligned}$$

Note that d_n^A has codomain $\ker d_{n-1}^A$ and thus $\overline{z_{n-1}} d_n^A = z_{n-1} d_n^A$. Similarly since h_n has codomain $\ker d_{n-1}^B$, we get $(\pi_{n-1} h_n)(c_n) = (\overline{\pi_{n-1}} h_n)(c_n)$. This gives us the desired commutativity. □

②: We can use the snake lemma on the diagram from ① to yield the following exact sequence:

$$\begin{array}{ccccccc} \ker d_n^A & \longrightarrow & \ker d_n^B & \longrightarrow & \ker d_n^C & \longrightarrow & \\ \text{coker } d_n^A & \longrightarrow & \text{coker } d_n^B & \longrightarrow & \text{coker } d_n^C & & \end{array}$$

But exactness of $A_\bullet$ and $C_\bullet$ implies $\text{coker } d_n^A = \text{coker } d_n^B = \phi$, and so we get an exact sequence $\phi \rightarrow \text{coker } d_n^B \rightarrow \phi$, which tells us $\text{coker } d_n^B = \ker d_{n-1}^B / \text{im } d_n^B = \phi$, as desired. □